

Heart rate following short-term beta-blocker titration predicts all-cause mortality in elderly chronic heart failure patients: insights from the CIBIS-ELD trial

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Aims

Beta-blockers (BBs) improve outcomes in heart failure. Results from the Cardiac Insufficiency Bisoprolol Study in Elderly (CIBIS-ELD) trial previously demonstrated the feasibility of heart rate, not maximum dose, as a treatment goal. In this pre-specified analysis, we investigated the prognostic value of achieved heart rate after BB optimization on long-term mortality.

Methods and results

Elderly heart failure patients from the CIBIS-ELD trial were invited to participate in a follow-up examination 4 years after the initial 12-week BB up-titration period. The relationship between all-cause mortality, BB dose, and heart rate after titration and potentially confounding clinical variables was analysed by multivariable Cox regression. In total, 728 patients (38% women; mean age 72.9 ± 5.4 years) were included. During a mean follow-up period of 45 ± 9 months, 134 patients (19%) died, thus accumulating 2268 patient-years at risk. There was no significant difference in baseline heart rate for survivors and non-survivors ($P = 0.19$). In models adjusting for age, sex, BB pre-treatment, ventricular function, heart rate, and NYHA class at baseline, a heart rate increase by 10 b.p.m. following up-titration was associated with a subsequent mortality hazard ratio of 1.19 (95% confidence interval 1.02–1.38, $P = 0.023$). The heart rate range with the lowest mortality and the fewest treatment-related adverse events was 55–64 b.p.m. The achieved BB dose was not associated with mortality risk.

Conclusion

The heart rate after up-titration, but not BB dose, predicted all-cause mortality risk in elderly patients with chronic heart failure. These patients should be titrated to resting heart rates between 55 and 64 b.p.m.

Keywords

Heart failure • Beta-blockers • Elderly • Heart rate

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Introduction

Beta-blockers are one of the cornerstones of heart failure pharmacotherapy because they reduce morbidity and mortality in these patients.¹ Current guidelines recommend titrating beta-blockers to a pre-defined target dose. However, in elderly patients, up-titration is frequently associated with challenging, dose-related side effects such as hypotension, bradycardia, dizziness, and sometimes worsening of concomitant COPD, predominantly with non-selective beta-blockers.^{2–4} As a consequence, recommended target doses can rarely be reached. In a recent analysis, we were able to demonstrate that among other dose-related effects, heart rate reduction was the major factor limiting beta-blocker up-titration to recommended doses.⁵

Though conceptually there are multiple effects of beta-blockade considered to be beneficial in heart failure (e.g. heart rate reduction, afterload reduction through blood pressure lowering, improvement of ventricular filling and coronary blood flow during diastole, decreased myocardial oxygen demand, decreased circulation of proinflammatory cytokines, and prevention of sudden cardiac death), recent data have indicated that heart rate reduction may be the single most important treatment effect; thus constituting an attractive treatment target.^{6–10} The prognostic value of heart rate in heart failure was originally highlighted in analyses of the MERIT-HF, CIBIS-II, and COMET trials, at that time with mixed results.^{11–13} More recently, a meta-analysis of 23 randomized, placebo-controlled beta-blocker trials in heart failure, including the aforementioned studies, demonstrated that heart rate reduction and not beta-blocker dose was associated with a significant survival benefit.¹⁴ Similarly, the investigators of the SHIFT trial concluded that for patients with resting heart rates of 70 b.p.m. and higher despite maximum tolerated beta-blocker doses 'heart-rate reduction with ivabradine reduces clinical events in heart failure in relation to the heart rate achieved'.¹⁵ In clinical practice this could mean that there is no 'one-size-fits-all' target beta-blocker dose as recommended by current guidelines, but rather a target heart rate that may be achieved by an individual beta-blocker dose. However, the lower limit for this optimal resting heart rate range has not yet been established.¹⁴

In this pre-specified long-term follow-up of the Cardiac Insufficiency Bisoprolol Study in Elderly (CIBIS-ELD) trial, we investigated the prognostic value of the achieved resting heart rate measured after the initial 12 weeks of beta-blocker up-titration on all-cause mortality in chronic heart failure patients. We hypothesized that this approach would allow the identification of an optimal heart rate range regarding the prevention of both clinical endpoints and adverse side effects.

Methods

Subjects and intervention

The study design and patient characteristics of the CIBIS-ELD trial, which compared the titration of bisoprolol and carvedilol to target doses with regard to beta-blocker tolerability, have been described in detail elsewhere.¹⁶ In summary, patients aged 65 years and older

with chronic stable heart failure with either reduced or preserved EF were randomized to either bisoprolol or carvedilol. At baseline, all patients were beta-blocker naïve or receiving $\leq 25\%$ of the target dose. Beta-blocker dose was titrated according to a standardized protocol every 2 weeks over a 12-week period, starting with 1.25 mg of bisoprolol daily or 3.125 mg of carvedilol twice daily. Target doses were 10 mg of bisoprolol daily or 25 mg of carvedilol twice daily. Investigators were allowed to adjust the dose level of the blinded treatment as needed. Adverse events were recorded at every visit; treatment-related events were defined according to a blinded committee. After the end of the main study, patients were invited to participate in an observational pre-specified follow-up. Patients with permanent pacemakers were excluded from this analysis.

The present follow-up analysis aims to investigate the relationships between achieved heart rate, clinical variables, achieved beta-blocker doses, and long-term mortality. The follow-up examinations took place at the CIBIS-ELD trial sites. In the case of non-attendance, a telephone follow-up was scheduled. Information on survival was also obtained via state registries.

Data collection and definitions

Data on demographics, medical history, NYHA class, ECG, echocardiography, 6 min walk test (6MWT), quality of life, and laboratory parameters were collected at baseline, after up-titration at 12 weeks, and at the follow-up examination (Table 1). Heart rate at baseline and after up-titration was measured via a 12-lead ECG after resting for 3 min in a supine position. Heart rate reduction was calculated as the difference between heart rate at baseline and at the end of the 12-week up-titration period. Glomerular filtration rate (GFR) was calculated using the MDRD (Modification of Diet in Renal Disease) formula.¹⁸

Data analysis

Baseline variables are presented as frequencies and percentages for binary variables, or means and standard deviations for quantitative variables (median and interquartile range for NT-proBNP). Data are presented overall and separately in the three groups according to achieved heart rates of 55–64, <55 , and >64 b.p.m. after up-titration. This is an observational trial and patients were not randomly allocated to heart rate target levels. Comparisons of all groups were performed by the χ^2 test or analysis of variance, respectively. When the overall test was significant ($P < 0.05$), groups were compared pairwise.

For beta-blocker dose, heart rate, systolic and diastolic blood pressure, 6MWT, NYHA class, and LVEF, the relationships between pre- and post-titration measurement and subsequent mortality were examined in univariable Cox regression and in two-variable Cox regression models including pre- and post-titration values at once.

In addition, the influence of multiple baseline characteristics on mortality was examined in a multivariable Cox regression model. Two models were computed, including heart rate as a continuous variable and in three categories (55–64, <55 , and >64 b.p.m.).

IBM SPSS version 21 was used as statistical software.

Ethics

The CIBIS-ELD trial (ISRCTN 34827306 at controlled-trials.com) and the present follow-up investigations were approved by the institutional ethical review boards of all participating centres. The study conforms with the principles outlined in the Declaration of Helsinki. All patients provided written informed consent prior to enrolment.

Table 1 Baseline characteristics

Variable	All patients (n = 728)	Heart rate (b.p.m.) at the end of beta-blocker up-titration			P-value
		55–64 (n = 281)	>64 (n = 348)	<55 (n = 99)	
Demographics					
Age (years), mean ± SD	72.9 ± 5.4	72.8 ± 5.3	72.6 ± 5.2	74.5 ± 5.4 ^{b,B}	0.007
Female sex, n (%)	278 (38)	106 (38)	137 (39)	35 (35)	0.753
Living alone, n (%)	284 (39)	115 (41)	131 (38)	38 (38)	0.697
Beta-blocker therapy					
Pre-treated with beta-blocker, n (%)	442 (61)	170 (61)	217 (62)	55 (56)	0.473
Randomized medication, n (%)					
Bisoprolol	360 (49)	143 (51)	160 (46)	57 (58)	0.105
Carvedilol	368 (51)	138 (49)	188 (54)	42 (42)	
Beta blocker dose after up-titration, mean ± SD	53 ± 36	52 ± 35	58 ± 37 ^a	42 ± 30 ^{a,C}	0.001
Clinical variables changed by beta-blocker therapy					
Heart rate on ECG (b.p.m.), mean ± SD					
Before up-titration	73 ± 14	69 ± 11	80 ± 14 ^c	62 ± 11 ^{c,C}	<0.001
End of up-titration	66 ± 13	60 ± 3	76 ± 11	51 ± 3	–
Change	–7 ± 13	–10 ± 11	–3 ± 14	–11 ± 11	–
Systolic blood pressure (mmHg), mean ± SD					
Before up-titration	138 ± 21	138 ± 20	136 ± 20	145 ± 27 ^{b,C}	0.002
End of up-titration	129 ± 19	129 ± 18	128 ± 18	133 ± 21 ^A	0.047
Change	–9 ± 21	–8 ± 20	–8 ± 21	–12 ± 23	0.344
Diastolic blood pressure (mmHg), mean ± SD					
Before up-titration	80 ± 12	80 ± 11	81 ± 12	80 ± 12	0.101
End of up-titration	76 ± 10	76 ± 10	77 ± 10 ^a	73 ± 11 ^{b,C}	<0.001
Change	–4 ± 12	–4 ± 11	–4 ± 13	–6 ± 13	0.210
NYHA class, mean ± SD					
Before up-titration	2.3 ± 0.5	2.3 ± 0.6	2.3 ± 0.5	2.2 ± 0.5	0.086
End of up-titration	2.0 ± 0.6	2.0 ± 0.6	2.0 ± 0.6	1.9 ± 0.5	0.071
Change	–0.3 ± 0.6	–0.3 ± 0.6	–0.3 ± 0.6	–0.3 ± 0.5	0.804
Six min walk distance (m), mean ± SD					
Before up-titration	325 ± 105	330 ± 101	310 ± 104 ^a	363 ± 108 ^{b,C}	<0.001
End of up-titration	340 ± 105	343 ± 100	326 ± 106 ^a	383 ± 105 ^{a,C}	<0.001
Change	+15 ± 65	+12 ± 63	+16 ± 66	+20 ± 64	0.539
LVEF (%), mean ± SD					
Before up-titration	43 ± 13	44 ± 14	40 ± 12 ^b	47 ± 13 ^{a,C}	<0.001
End of up-titration	45 ± 13	46 ± 13	44 ± 12 ^b	49 ± 12 ^C	<0.001
Change	+3 ± 7	+2 ± 7	+3 ± 7	+2 ± 8	0.191
Co-morbidities, risk factors, and clinical variables					
Peripheral oedema, n (%)	79 (11)	25 (9)	42 (12)	12 (12)	0.408
Hospitalization for heart failure during past year, n (%)	255 (35)	99 (35)	124 (36)	32 (32)	0.827
Atrial fibrillation, n (%)	140 (19)	29 (10)	107 (31) ^c	4 (4) ^C	<0.001
Coronary heart disease, n (%)	426 (59)	166 (59)	199 (57)	61 (62)	0.711
Body mass index (kg/m ²), mean ± SD	27.9 ± 4.8	27.8 ± 5.0	28.0 ± 4.8	27.9 ± 4.2	0.933
Diabetes mellitus, n (%)	191 (26)	54 (19)	115 (33) ^c	22 (22) ^A	<0.001
Current smoker, n (%)	65 (9)	28 (10)	33 (10)	4 (4)	0.203
COPD, n (%)	53 (7)	16 (6)	29 (8)	8 (8)	0.429
FEV ₁ (mL), mean ± SD	2181 ± 651	2266 ± 652	2097 ± 641 ^b	2234 ± 654	0.004
History of depression, n (%)	59 (8)	20 (7)	28 (8)	11 (11)	0.460
Current PHQ-9 sum score, mean ± SD	5.9 ± 4.9	5.5 ± 4.9	6.1 ± 4.8	6.4 ± 5.4	0.246

Table 1 Continued

Variable	All patients (n = 728)	Heart rate (b.p.m.) at the end of beta-blocker up-titration			P-value
		55–64 (n = 281)	>64 (n = 348)	<55 (n = 99)	
Laboratory measurements					
Serum sodium (mmol/L), mean ± SD	141 ± 4	141 ± 4	141 ± 3	141 ± 4	0.718
Serum potassium (mmol/L), mean ± SD	4.4 ± 0.5	4.4 ± 0.5	4.4 ± 0.5	4.4 ± 0.5	0.968
Haemoglobin (g/dL), mean ± SD	13.5 ± 1.5	13.6 ± 1.5	13.5 ± 1.5	13.6 ± 1.5	0.904
eGFR (mL/min/1.73 m ² BSA), mean ± SD	64 ± 18	65 ± 19	64 ± 18	64 ± 15	0.813
Uric acid (mg/dL), mean ± SD	5.9 ± 1.8	5.8 ± 1.6	6.0 ± 2.0	6.1 ± 1.6	0.192
Total cholesterol (mg/dL), mean ± SD	205 ± 48	208 ± 48	204 ± 49	198 ± 46	0.212
NT-proBNP (ng/L), median (interquartile range)	612 (265–1431)	496 (241–1204)	883 ^c (343–1782)	408 ^C (200–952)	<0.001
Concomitant therapy					
ACE inhibitor and/or ARB, n (%)	616 (85)	239 (85)	289 (83)	88 (89)	0.356
Mineralocorticoid receptor antagonist, n (%)	224 (31)	83 (30)	116 (33)	25 (25)	0.262
Diuretic, n (%)	552 (72)	191 (68)	259 (74)	72 (73)	0.198
Cardiac glycoside, n (%)	94 (13)	24 (9)	63 (18) ^c	7 (7) ^A	<0.001
Statin, n (%)	296 (41)	111 (40)	139 (40)	46 (46)	0.448
Antiplatelet therapy, n (%)	496 (68)	197 (70)	234 (67)	65 (66)	0.634
Anticoagulant, n (%)	164 (23)	49 (17)	101 (29) ^c	14 (14) ^B	<0.001
Antiarrhythmic drugs, n (%)	72 (10)	35 (12)	31 (9)	6 (6)	0.136
Antidepressant, n (%)	25 (3)	10 (4)	12 (3)	3 (3)	0.970

BSA, body surface area; eGFR, estimated glomerular filtration rate; FEV₁, forced expiratory volume in 1 s; PHQ, Patient Health Questionnaire.

^aP < 0.05; ^bP < 0.01; ^cP < 0.001 vs. 55–64 b.p.m. ^AP < 0.05; ^BP < 0.01; ^CP < 0.001 vs. >64 b.p.m.

Results

Sample characteristics

Of the 751 eligible participants, 728 patients (38% women; mean age 72.9 ± 5.4 years) were included in this analysis, thus accumulating 2268 patient-years at risk (mean follow-up 44.6 ± 9.2 months; Figure 1). During that period, 134 patients (19%) died. Baseline characteristics according to heart rate tertiles following beta-blocker titration are shown in Table 1. On average, patients in the lowest tertile (<55 b.p.m.) were 2 years older than subjects with heart rates ≥55 b.p.m. The highest tertile (>64 b.p.m.) had higher baseline heart rates, a shorter six min walk test distance, lower LVEF, and higher NT-proBNP levels (*P* < 0.001, Table 1).

Beta-blocker pre-treatment, achieved dose, and mortality

There were no differences in the proportion of patients pre-treated with beta-blockers between the three post-titration heart rate tertiles. One-third of all patients reached guideline-recommended target doses. Of note, after the up-titration period, patients with lower heart rates received lower mean beta-blocker doses (42 ± 30% of the target dose in the lowest tertile vs. 58 ± 37% in the highest tertile; Table 1). In the multivariable regression analysis, neither beta-blocker agent (either bisoprolol or carvedilol) nor the achieved dose predicted mortality.

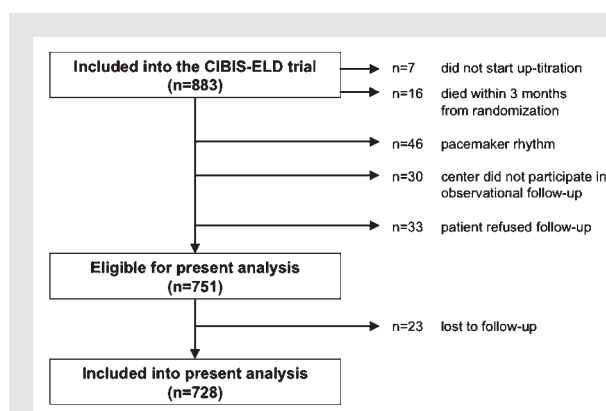


Figure 1 Flowchart of patient selection for analysis.

Heart rate and adverse events

Half of the patients experienced adverse events during up-titration. The most common adverse event was worsening heart failure (21.6%) followed by bradycardia (14.1%), new atrioventricular block (9.6%), hypotension (9.1%), fatigue (8.4%), vertigo (7.3%), and worsening of COPD (6.7%). In a multivariable regression model that included age, sex, prior beta-blocker treatment, ventricular function, baseline heart rate, and baseline NYHA class, a heart rate between 55 and 64 b.p.m. achieved after beta-blocker up-titration was associated with the lowest rate of treatment-related events [odds ratio (OR) 0.04, 95% confidence

Table 2 Multivariable Cox regression to predict mortality

Variable	Continuous end-titration heart rate		Categorical end-titration heart rate	
	Hazard ratio (95% CI)	P-value	Hazard ratio (95% CI)	P-value
Heart rate				
Before BB up-titration (per +10 b.p.m.)	0.96 (0.83–1.11)	0.605	0.96 (0.83–1.11)	0.559
After BB up-titration (per +10 b.p.m.)	1.19 (1.02–1.38)	0.023	–	–
After BB up-titration (>64 vs 55–64 bpm)	–	–	1.84 (1.20–2.80)	0.005
After BB up-titration (<55 vs. 55–64 bpm)	–	–	1.09 (0.58–2.07)	0.787
Beta-blocker				
Pre-treatment	0.42 (0.29–0.62)	<0.001	0.41 (0.28–0.61)	<0.001
Up-titrated agent (carvedilol vs. bisoprolol)	1.20 (0.85–1.70)	0.292	1.21 (0.86–1.72)	0.277
Equivalence dose after up-titration				
≤25% of recommended target dose	1.34 (0.83–2.14)	0.228	1.41 (0.88–2.25)	0.158
50% of recommended target dose	1.44 (0.87–2.39)	0.159	1.45 (0.88–2.41)	0.148
Full recommended target dose	1.00 (reference)	–	1.00 (reference)	–
Covariables (values after up-titration)				
Age (per +10 years)	2.17 (1.58–2.99)	<0.001	2.08 (1.51–2.84)	<0.001
Female sex	0.71 (0.46–1.08)	0.109	0.68 (0.45–1.05)	0.084
Coronary heart disease	1.70 (1.13–2.56)	0.011	1.69 (1.13–2.53)	0.011
NYHA class (per class higher)	1.01 (0.72–1.41)	0.955	1.03 (0.73–1.44)	0.870
Six min walk distance (per +100 m)	0.84 (0.70–1.00)	0.053	0.84 (0.70–1.02)	0.073
LVEF (per +10%)	0.72 (0.60–0.86)	<0.001	0.73 (0.61–0.87)	<0.001
Heart failure hospitalization in past year	1.34 (0.95–1.90)	0.096	1.32 (0.93–1.88)	0.117
Haemoglobin (per +1 g/dL)	0.84 (0.75–0.94)	0.003	0.83 (0.74–0.94)	0.003
eGFR (per +10 mL/min/1.73 m ²)	1.03 (0.93–1.14)	0.615	1.03 (0.93–1.14)	0.567
NT-proBNP (per 10-fold increase)	2.08 (1.40–3.09)	<0.001	2.16 (1.45–3.21)	<0.001

BB, beta-blocker; CI, confidence interval; eGFR, estimated glomerular filtration rate.

interval (CI) 0.004–0.39, $P = 0.006$). In contrast, a heart rate <55 b.p.m. following up-titration was associated with the highest rate of adverse events (OR 2.58, 95% CI 1.53–4.48, $P < 0.001$).

Heart rate and mortality

Prior to beta-blocker up-titration there was no significant difference in heart rate observed for survivors and non-survivors (72.6 ± 13.9 b.p.m. vs. 74.3 ± 14.1 b.p.m., $P = 0.19$). After up-titration, heart rates were lowered to 69 ± 13 b.p.m. in non-survivors vs. 66 ± 12 b.p.m. in survivors, respectively (difference adjusted for baseline heart rate, 2.7 b.p.m., 95% CI 0.7–4.7, $P = 0.007$). In a multiple Cox regression adjusting for age, sex, beta-blocker pre-treatment, ventricular function, heart rate, and NYHA class at baseline, each heart rate increase of 10 b.p.m. following up-titration was associated with a subsequent mortality hazard ratio (HR) of 1.19 (95% CI 1.02–1.38, $P = 0.023$; Table 2). A heart rate ≤ 64 b.p.m. was associated with lower mortality compared with the highest heart rate category >64 b.p.m. (HR 1.84, 95% CI 1.20–2.88, $P = 0.005$; Table 2; Figure 2). Heart rates <55 b.p.m. after up-titration were no more beneficial in regard to survival than heart rates in the range of 55–64 b.p.m. (HR 1.09, CI 0.58–2.07, $P = 0.787$; Table 2; Figure 2). No statistical difference ($P < 0.05$) was observed in respect to the effect of heart rate on survival in the subgroups defined by beta-blocker

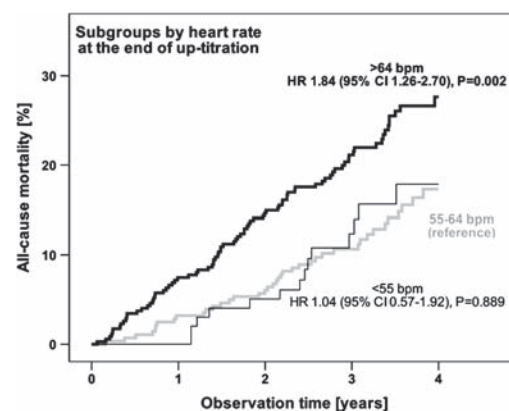


Figure 2 Kaplan–Meier curves showing the effects of heart rate following beta-blocker up-titration on mortality. CI, confidence interval; HR, hazard ratio.

pre-treatment (no/yes), age (below/above median), and LV function (reduced/preserved) (Figure 3).

Discussion

Achieved resting heart rate between 55 and 64 b.p.m. following a 12-week beta-blocker up-titration period was associated with

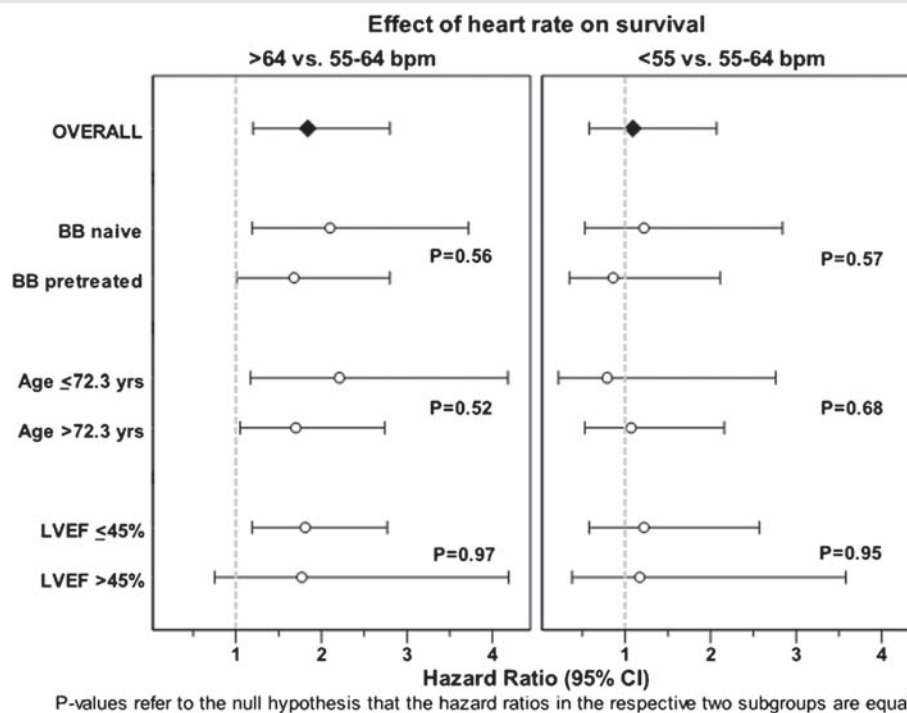


Figure 3 Relationship between heart rate after titration and mortality overall and in the subgroups defined by beta-blocker (BB) pre-treatment (no/yes), age (below/above median), and LV function (reduced/preserved).

lower long-term mortality risk and the lowest frequency of adverse events. For every 10 b.p.m. increase in heart rate, a corresponding 19% increase was seen in all-cause mortality after adjusting for age, sex, beta-blocker pre-treatment, ventricular function, heart rate, and NYHA class at baseline.

In a 2008 meta-analysis of 26 placebo-controlled heart failure beta-blocker trials that reported on heart rate and systolic function and nine further trials with available mortality data, a strong correlation was found between heart rate reduction, changes in the LVEF, annualized all-cause mortality, and achieved heart rate.¹⁷ The benefit of heart rate reduction on all-cause mortality was later quantified in a meta-analysis of 23 placebo-controlled beta-blocker trials of systolic heart failure.¹⁴ Interestingly, a survival benefit was associated with the type of beta-blocker agent used (benefits observed only for metoprolol, carvedilol, and bisoprolol) but not with beta-blocker dose. Based on the aggregate data from 19 209 patients, an 18% reduction in mortality was seen for every 5 b.p.m. reduction in heart rate (95% CI 6–29%). However, no inflection point for the optimal heart rate range could be identified in the meta-analysis. In our study in elderly patients, a 19% increase was seen in all-cause mortality for every 10 b.p.m. increase in heart rate. Interestingly, the relationship between heart rate after titration and mortality in the subgroups of beta-blocker pre-treatment (no/yes), age (below/above median), and LV function (reduced/preserved) was similar. These findings suggest a beneficial effect of heart rate reduction in patients with preserved EF. Further, the increased mortality risk with a heart rate >64 b.p.m. as

compared with 55–64 b.p.m. in Cox regression was of the same magnitude as the increased risk associated with age increased by 8.3 years, a 7.9-fold increase NT-proBNP, or a drop of haemoglobin by 3.3 g/dL.

We were able to identify a resting heart rate range from 55 to 64 b.p.m. following beta-blocker treatment that corresponded to lower long-term mortality and the fewest treatment-related adverse events. Thus, this range seems a reasonable target for beta-blocker titration. Our data suggest that conceivably beta-blocker dosing beyond established beta-blocker target doses may be safe and necessary in order to achieve optimal heart rate control in patients with treatment-resistant elevated heart rates at common beta-blocker doses.

Our data are supported by the findings of a recent large single-centre observational study by Cullington *et al.*, which enrolled 654 patients with sinus rhythm and chronic stable heart failure due to LV systolic dysfunction at a heart failure clinic.¹⁰ In this study, beta-blocker treatment had been initiated in 58% of patients prior to enrolment. Over a period of 4 months, patients were then up-titrated to target doses, after which 82% of patients received beta-blockers; of note, only 18% achieved 'high-dose beta-blockade' defined as >66% of target dose. In a multivariable model, resting heart rate at 4 months and beta-blocker use vs. no beta-blocker treatment were independent predictors of survival. However, heart rate reduction (the difference between heart rate at baseline and at 4 months) and beta-blocker dose did not predict survival. The authors concluded that the optimal heart rate range

may lie between 58 and 64 b.p.m., although the Kaplan–Meier curve showed no statistically significant difference between the two lowest heart rate quartiles (≤ 57 and 58–64 b.p.m.), as it was assumed that symptomatic bradycardia would lead to higher mortality at rates < 58 b.p.m. Furthermore, they speculated that the benefit of heart rate reduction may be independent of the rate-lowering agent used. This question might be answered by an adequately powered trial comparing head-to-head the prognostic benefit mediated by beta-blockade vs. other rate-limiting agents, such as I_f channel sinus node inhibition with ivabradine. Despite the similar findings, several key differences in sample composition between the Cullington study and the present study should be mentioned, due to potential implications on generalizability and application of these findings to daily clinical practice. Although originating from a clinical trial, our sample is more representative of ‘real-world’ heart failure care. Despite similar age, the present study included more women and more beta-blocker-naïve patients, and had a longer follow-up period compared with the study of Cullington. On average, our patients were more symptomatic; they were able to reach higher beta-blocker doses and received more guideline-recommended care (such as mineralocorticoid receptor antagonists in systolic heart failure).

Over the 12-week intervention period, only one-third of patients could be up-titrated to target doses. However, $> 50\%$ achieved heart rates < 65 b.p.m., which were associated with a significant survival benefit over a 4-year follow-up period, during which patients received no further specialized care. Thus, even a short supervised forced titration of beta-blockers, for instance in a specialized heart failure clinic, is feasible and advisable for improving outcomes in heart failure patients. A representative survey study of heart failure awareness and management among cardiologists, internists, geriatricians, and primary care physicians in Europe found that the self-reported adherence to guideline-recommended care was highest among patients treated by cardiologists. One-third of primary care physicians, 25% of internists and geriatrics, and 11% of cardiologists responded that they would not consider starting beta-blockers in patients with mild heart failure, who were already on an ACE inhibitor and a diuretic.¹⁹ A review of 18 randomized studies of heart failure clinics with specialized nursing vs. usual care demonstrated higher rates of guideline-recommended care and a significant reduction in readmission rates and mortality in the intervention groups.²⁰ In addition to specialized care at heart failure clinics, continuing professional education programmes that target primary care physicians and general internists should incorporate heart rate goals into their heart failure curriculum. Until prospective randomized rate-ranging studies in chronic heart failure have established target heart rates, we consider a heart rate between 55 and 65 b.p.m. a pragmatic treatment goal in these patients.

Limitations

Due to the observational nature of this pre-specified analysis, cautious interpretation of our findings is warranted. The three heart rate groups showed significant variation in baseline demographic and clinical characteristics. Despite adjustment for these variables via multivariable regression analyses, we cannot rule out

the effect of unmeasured confounding. Furthermore, we can make no recommendation as to which agent is to be preferred for lowering heart rate due to incomplete data about beta-blocker treatment during the 4-year follow-up period. Patients were not randomly allocated to heart rate target levels. Randomized controlled trials are required to prove that an optimal heart rate target should be between 55 and 64 b.p.m. Our population was slightly younger compared with the real-life elderly population which actually include patients over 75 years old.

Conclusion

In this study we demonstrated that achieved heart rate ≤ 64 b.p.m. after short-term beta-blocker up-titration was associated with long-standing lower mortality in elderly heart failure patients. Rather than targeting a maximal beta-blocker dose, physicians should consider titrating beta-blockers to reach a resting heart rate between 55 and 64 b.p.m. At this moment, we suggest that heart failure patients should be referred to heart failure specialists for supervised beta-blocker up-titration.

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